

Review Note: Aggregation from Firm-Level to Sector-Level Extensive Margin (Levels)

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1 Setup and Notation

This note reviews the claim in Section 2.1 (“Extensive margin (levels)”) that the firm-level and sector-level extensive-margin regressions recover the same $\lambda_\ell^{\text{ext}}$ coefficient when every firm receives equal weight. We verify the aggregation algebra, identify the exact conditions under which the claim holds, and clarify what happens when the firm-level regression uses employment weights instead.

Firm-level regression. For a generic cell $c = (j, m, e)$ containing the pre-election firm set $\mathcal{F}_{jme}^{\text{pre}}$ with $N_c \equiv |\mathcal{F}_{jme}^{\text{pre}}|$ firms, and for each post-election year $t \in \{e + 1, \dots, e + 4\}$, the paper specifies

$$\mathbf{1}(\text{BNDES}_{fmt} > 0) = \sum_{\ell} \lambda_{\ell}^{\text{ext}} \text{FA}_{fmt}^{\ell} + \gamma_f + \alpha_{mt} + u_{fmt}. \quad (1)$$

Sector-level aggregation. The paper defines

$$H_{jmt}^{\text{pre}} \equiv \frac{1}{N_c} \sum_{f \in \mathcal{F}_{jme}^{\text{pre}}} \mathbf{1}(\text{BNDES}_{fmt} > 0), \quad (2)$$

$$\overline{\text{FA}}_{jmt}^{\ell, \text{pre}} \equiv \frac{1}{N_c} \sum_{f \in \mathcal{F}_{jme}^{\text{pre}}} \text{FA}_{fmt}^{\ell}, \quad (3)$$

and claims that averaging (1) across the pre-election firm base yields

$$H_{jmt}^{\text{pre}} = \sum_{\ell} \lambda_{\ell}^{\text{ext}} \overline{\text{FA}}_{jmt}^{\ell, \text{pre}} + \bar{\gamma}_{jme} + \alpha_{mt} + \epsilon_{jmt}, \quad (4)$$

with $\bar{\gamma}_{jme} = N_c^{-1} \sum_{f \in \mathcal{F}_{jme}^{\text{pre}}} \gamma_f$ and $\epsilon_{jmt} = N_c^{-1} \sum_{f \in \mathcal{F}_{jme}^{\text{pre}}} u_{fmt}$.

2 Verification of the Averaging Step

Proposition 1 (Algebraic correctness of the averaging). *If equation (1) holds for every $f \in \mathcal{F}_{jme}^{\text{pre}}$, then summing both sides over f and dividing by N_c yields equation (4) exactly, with the same $\lambda_{\ell}^{\text{ext}}$.*

Proof. Sum (1) over $f \in \mathcal{F}_{jme}^{\text{pre}}$:

$$\sum_f \mathbf{1}(\text{BNDES}_{fmt} > 0) = \sum_{\ell} \lambda_{\ell}^{\text{ext}} \sum_f \text{FA}_{fmt}^{\ell} + \sum_f \gamma_f + N_c \alpha_{mt} + \sum_f u_{fmt}.$$

Divide both sides by N_c :

$$\underbrace{\frac{1}{N_c} \sum_f \mathbf{1}(\text{BNDES}_{fmt} > 0)}_{= H_{jmt}^{\text{pre}}} = \sum_{\ell} \lambda_{\ell}^{\text{ext}} \underbrace{\frac{1}{N_c} \sum_f \text{FA}_{fmt}^{\ell}}_{= \overline{\text{FA}}_{jmt}^{\ell, \text{pre}}} + \underbrace{\frac{1}{N_c} \sum_f \gamma_f + \alpha_{mt}}_{= \bar{\gamma}_{jme}} + \underbrace{\frac{1}{N_c} \sum_f u_{fmt}}_{= \epsilon_{jmt}}.$$

This is exactly (4). The $\lambda_{\ell}^{\text{ext}}$ coefficients pass through unchanged because averaging is a linear operation and the coefficients are common to all firms. $\square$

Remark 1 (The α_{mt} term). The municipality $\times$ year fixed effect α_{mt} does not carry a firm subscript, so $\frac{1}{N_c} \sum_f \alpha_{mt} = \alpha_{mt}$. This is correct in the paper.

3 Under What Conditions Are the Estimated Coefficients Identical?

The algebraic identity above shows that the *population* equation aggregates exactly. However, the *estimated* $\hat{\lambda}$ from OLS on the firm-level data and from OLS on the cell-averaged data need not be numerically equal. The key issue is the weighting implicit in each estimator.

3.1 Case 1: Unweighted firm-level OLS

In the unweighted firm-level regression (1), every firm-observation (f, t) receives weight 1. By the Frisch–Waugh–Lovell (FWL) theorem, after absorbing fixed effects (γ_f, α_{mt}) , OLS minimizes

$$\sum_c \sum_{f \in \mathcal{F}_{jme}^{\text{pre}}} \left(\tilde{Y}_{fmt} - \sum_{\ell} \lambda_{\ell} \tilde{X}_{fmt}^{\ell} \right)^2,$$

where tildes denote residuals after the FE projection. Each cell c contributes N_c terms to the sum.

Now consider running OLS on the cell-averaged equation (4), where each cell (j, m, t) is one observation. Each cell gets weight 1, so OLS minimizes

$$\sum_c \left(\tilde{Y}_c - \sum_{\ell} \lambda_{\ell} \tilde{X}_c^{\ell} \right)^2.$$

Here each cell contributes exactly 1 term regardless of how many firms it contains.

Proposition 2 (Equal coefficients iff cell-size weights). Let $\hat{\lambda}^{\text{firm}}$ be the OLS estimator from the unweighted firm-level regression (1), and let $\hat{\lambda}^{\text{sector}}$ be the OLS estimator from the cell-averaged regression (4). Then:

$$\hat{\lambda}^{\text{firm}} = \hat{\lambda}^{\text{sector}} \iff \text{the sector-level regression weights each cell by } N_c.$$

Proof. Consider the firm-level residuals after FE projection. For firm f in cell c , the within-cell residual is $\tilde{Y}_{fmt} = Y_{fmt} - \bar{Y}_c - \bar{Y}_f + \bar{Y}$, and similarly for X .

We can decompose the firm-level sum of squares into a *between-cell* component and a *within-cell* component. The between-cell normal equation from the firm-level OLS is

$$\sum_c N_c \tilde{X}_c' (\tilde{Y}_c - \tilde{X}_c \lambda) = 0,$$

i.e., the cell means $\tilde{X}_c, \tilde{Y}_c$ enter with weight N_c . An unweighted regression on cell means weights every cell equally. The two coincide if and only if N_c is used as an analytic weight in the sector-level regression.

Intuitively: if cell A has 10 firms and cell B has 1 firm, the firm-level regression lets cell A contribute 10 times the moment conditions. The cell-level regression with unit weights treats them equally. Only weighting by N_c restores the equivalence. $\square$

3.2 Case 2: Employment-weighted firm-level OLS

The primary specification in the paper uses analytic regression weights $n_{\text{emp},ft}$. The firm-level estimator then solves

$$\sum_c \sum_{f \in \mathcal{F}_{jme}^{\text{pre}}} n_{\text{emp},ft} \tilde{X}_{fmt}^{\ell} (\tilde{Y}_{fmt} - \sum_{\ell} \lambda_{\ell} \tilde{X}_{fmt}^{\ell}) = 0.$$

In this case, even weighting the sector-level regression by N_c does *not* replicate the firm-level estimator, because within each cell employment-weighted averaging gives more influence to larger firms.

To recover the employment-weighted firm-level coefficient at the sector level, one would need to:

1. Define employment-weighted cell averages:

$$H_{jmt}^{\text{emp}} = \frac{\sum_f n_{\text{emp},ft} \mathbf{1}(\text{BNDES}_{fmt} > 0)}{\sum_f n_{\text{emp},ft}}, \quad \overline{\text{FA}}_{jmt}^{\ell, \text{emp}} = \frac{\sum_f n_{\text{emp},ft} \text{FA}_{fmt}^{\ell}}{\sum_f n_{\text{emp},ft}},$$

where sums run over $f \in \mathcal{F}_{jme}^{\text{pre}}$.

2. Weight the sector-level regression by total cell employment $E_c = \sum_{f \in \mathcal{F}_{jme}^{\text{pre}}} n_{\text{emp},ft}$.

Only then would $\hat{\lambda}_{\text{sector}} = \hat{\lambda}_{\text{firm}}$.

The paper’s current sector-level aggregation uses *equal-firm* (simple average) weights, $1/N_c$, which corresponds to the *unweighted* firm-level estimator, not the employment-weighted one. This is not an error — it is a different estimand. The equal-firm sector average asks about the average firm; the employment-weighted version asks about the average worker’s firm.

4 The “Most Cells Have 1–2 Firms” Argument

The user raises the following intuition: since most municipality-sector cells contain only 1 or 2 firms, cells with more firms should receive proportionally more weight in the sector regression so that each *firm* carries the same influence. This is exactly Proposition 2.

Remark 2 (Practical magnitude of the discrepancy). *If most cells have $N_c = 1$, then N_c -weighting barely changes anything—for singleton cells, the cell average is the firm observation. The discrepancy comes entirely from the few cells with $N_c > 1$. In those cells:*

- The unweighted sector regression treats the cell as one observation (the average of, say, 5 firms gets weight 1).
- The N_c -weighted sector regression gives that cell weight 5.
- The firm-level regression gives each of the 5 firms weight 1, for a total cell contribution of 5.

So the N_c -weighted sector regression reproduces the firm-level OLS exactly, while the unweighted version under-represents the larger cells.

Whether the discrepancy matters empirically depends on:

1. The fraction of observations in multi-firm cells.
2. Whether the within-cell variation in FA_{fmt}^{ℓ} is large relative to between-cell variation.

If most identifying variation is between cells rather than within cells, the two estimators will be close even without N_c -weighting.

5 Summary of the Claim in Line 223

The paper states (line 223):

“In theory, the $\lambda_{\ell}^{\text{ext}}$ coefficients from both the firm-level and the sector-level should be the same, at least if each firm has equal weight in both cases.”

Assessment. This statement is **correct but incomplete**. Specifically:

1. **The algebraic averaging is correct.** Summing the firm-level equation over $\mathcal{F}_{jme}^{\text{pre}}$ and dividing by N_c yields equation (4) with the same population $\lambda_{\ell}^{\text{ext}}$. No errors in the math.
2. **“Equal weight” must be enforced in both regressions simultaneously.** The condition “each firm has equal weight” means:
 - The firm-level regression is *unweighted* (each firm-year observation gets weight 1).

- The sector-level regression uses analytic weights equal to the cell size $N_c = |\mathcal{F}_{jme}^{\text{pre}}|$.

If the sector-level regression is run unweighted (each cell gets weight 1), then firms in large cells are down-weighted relative to the firm-level regression, and $\hat{\lambda}^{\text{sector}} \neq \hat{\lambda}^{\text{firm}}$ in general.

3. **The primary firm-level specification uses employment weights.** With employment weights at the firm level, equal-firm averaging at the sector level does not recover the same estimand. To match, one would need employment-weighted sector averages and total-employment cell weights (see Section 3.2 above). The paper wisely does not claim the primary (employment-weighted) firm-level and sector-level coefficients are the same; the claim is conditional on “equal weight.”
4. **Additional assumption: same sample.** The aggregation identity requires that the sector-level regression uses *exactly* the same firm set $\mathcal{F}_{jme}^{\text{pre}}$ as the firm-level regression. If the sector regression drops cells that have no variation in FA (e.g., cells where all firms have zero affiliation), or if the firm panel has missing observations for some firms in some years, the samples diverge and the coefficients will differ even with correct weighting.
5. **Additional assumption: same fixed-effect structure.** The firm-level regression absorbs γ_f (firm FE) and α_{mt} (municipality×year FE). The cell-averaged regression absorbs $\bar{\gamma}_{jme}$ (cell-level FE, which is a municipality×sector×ele cycle FE) and α_{mt} . These are algebraically consistent *only if* the cell-averaged regression is specified with the correct grouped FE. In practice, the sector-level estimation in the paper uses γ_{jm} (municipality×sector FE) and α_{jt} (sector×year FE), which is a different FE structure from $\bar{\gamma}_{jme} + \alpha_{mt}$. This means the sector-level estimator is *not* a mechanical aggregation of the firm-level one; it is a complementary specification with its own identification.

6 Recommendation

The claim on line 223 is mathematically sound for the population equation. To make the paper’s discussion precise, consider strengthening it to:

“In theory, if (i) the firm-level regression is unweighted, (ii) the sector-level regression weights each cell by N_{jme}^{pre} , (iii) both regressions use the same sample of firms, and (iv) the sector-level fixed effects nest the averaged firm fixed effects, then the estimated $\lambda_{\ell}^{\text{ext}}$ from the two regressions are numerically identical. In practice, the sector-level specification uses a different fixed-effect structure ($\gamma_{jm} + \alpha_{jt}$ rather than $\bar{\gamma}_{jme} + \alpha_{mt}$), so the two estimators target related but distinct causal parameters. Comparing them is informative about the robustness of the political-lending channel across levels of aggregation.”

7 Relation to the Sector-Level Instrument Z_{jmt}^{ℓ}

A separate but related point: the sector-level instrument defined in Section 2.3 of the paper is

$$Z_{jmt}^{\ell} = \sum_p w_{jmp,t}^{\ell} \text{Align}_{mpt}^{\ell},$$

where $w_{jmp,t}^{\ell}$ is the *owner-count-weighted* average of firm-level exposures:

$$w_{jmp,t}^{\ell} = \sum_{f \in \mathcal{F}(j,m)} \frac{L_{f,t}}{N_{jm,t}^{\ell}} \omega_{fp,t}^{\ell}.$$

This uses total-owner-count weighting, not equal-firm weighting. The “aggregation link” paragraph (lines 330–347) confirms that Z_{jmt}^{ℓ} is the *owner-count-weighted* average of FA_{fmt}^{ℓ} :

$$Z_{jmt}^{\ell} = \sum_{f \in \mathcal{F}_{jme}^{\text{pre}}} \frac{L_{f,t}}{\sum_{f'} L_{f',t}} \text{FA}_{fmt}^{\ell}.$$

This is distinct from the *equal-firm average* $\overline{\text{FA}}_{jmt}^{\ell, \text{pre}} = N_c^{-1} \sum_f \text{FA}_{fmt}^{\ell}$ used in the extensive-margin aggregation. The two objects would coincide only if all firms in the cell have the same number of owners.

Therefore, the paper is internally consistent: the sector-level first stage (Section 2.3) uses Z_{jmt}^{ℓ} as an instrument for s_{jmt} , while Section 2.1's aggregation discussion uses $\overline{\text{FA}}_{jmt}^{\ell, \text{pre}}$ as the sector counterpart of the firm-level LPM. These are different objects serving different purposes, and the paper should not conflate them.